

Koch curve - v0 - review 0

This is an outstanding piece of mathematical writing. You have successfully thread a very difficult needle: making the geometric intuition accessible to a pre-graduate student while providing the rigorous functional, variational, and spectral scaffolding demanded by a working analyst.

The integration of analytic proofs with exact, to-the-machine-precision numerical verifications (and the inclusion of the Python code that generated them) is a masterstroke. It bridges the gap between abstract fractal geometry and computational mathematics beautifully.

Since you requested **severe scrutiny**, I have reviewed the mathematical claims, proofs, historical context, and pedagogical flow with a magnifying glass. I found the essay to be essentially flawless in its core math, but I have a few nuanced critiques, theoretical suggestions, and minor semantic tweaks that will elevate this from "excellent" to "definitive."

Here is my section-by-section scrutiny.

1. Strengths to Preserve (Do not change these)

- **The p -variation proof (Section 6.3):** Your demonstration that $V_{p^*}(\varphi) = 1$ is exceptionally slick. Using the general $C^{0,\alpha}$ partition bound to prove it is *finite*, and then using the vertex partition to lock the supremum exactly at 1, is elegant and rigorous.
- **The Fourier exact recursion (Section 7.1):** Deriving the exact Fourier coefficients from the Hutchinson functional equation is rare to see written out so cleanly. Proving the Hadamard-gap identity $c_{4^n} = c_1/4^n$ purely algebraically is the crown jewel of this essay.
- **The "Dual-Audience" Tone:** The use of "Remark" blocks (like the one connecting p -variation to rough paths) is the perfect way to feed the working analyst without alienating the undergraduate.

2. "Severe Scrutiny" Critiques & Suggestions

A. Clarification on "Arc-Length Fraction" (Section 2.1) You write that P_n is parameterized by the "arc-length fraction $t \in [0, 1]$ ". For the polygonal approximations P_n , this is completely accurate, as all 4^n segments have equal Euclidean length. However, it is worth adding a half-sentence noting that in the limit φ , this parameter t transitions from representing arc-length to representing the **natural mass distribution (or Hausdorff measure)** of the fractal, since the concept of arc-length vanishes.

B. The Symmetry/Fourier Connection (Section 7.2) You note: "...so $\varphi(1-t) = 1 - \overline{\varphi(t)}$... This forces every c_k to be purely imaginary." Because this essay targets analysts, I highly recommend including the 1.5 lines of algebra that prove this. It is too elegant to leave as an exercise. *Sketch:* Let $u = 1 - t$. Then $c_k = \int_0^1 \varphi(1-u)e^{-2\pi i k(1-u)} du$. Since $e^{-2\pi i k} = 1$, this becomes $\int_0^1 (1 - \overline{\varphi(u)})e^{2\pi i k u} du$. The integral of $e^{2\pi i k u}$ vanishes for $k \geq 1$, leaving $c_k = -\int_0^1 \varphi(u)e^{-2\pi i k u} du = -\overline{c_k}$. Therefore, c_k is purely imaginary.

C. The Divergence of $\sum |c_k|$ vs. Weierstrass (Section 7.2) You correctly point out that $\sum |c_k| = \infty$, relying on the envelope and numerical evidence $C \log N$. You should contrast this explicitly with the Weierstrass function mentioned in your introduction. The classical Weierstrass function $W(x) = \sum a^n \cos(b^n \pi x)$ **is** absolutely convergent (since $\sum a^n < \infty$). The Koch curve is arguably more fascinating because its lacunary subsequence $\sum |c_{4^n}|$ converges, but the "noise" (the non-lacunary coefficients filling the gaps) forces the total sum to diverge, meaning the Koch parametrization does *not* belong to the

Wiener algebra $A(\mathbb{T})$. Pointing out this specific divergence from the Weierstrass behavior adds deep analytic value.

D. The Hölder Upper Bound Logic (Section 6.1) Your proof of the Hölder upper bound is brilliant, but there is a tiny, implicit geometric step that an eagle-eyed reviewer will catch. You write: $|P_n(t) - P_n(s)| \leq 2 \cdot 3^{-n}$. You should explicitly state *why*: "Because $|t - s| \leq 4^{-n}$, the parameters t and s can span at most **two adjacent** level- n segments. The path distance along the polygon is thus at most $2 \cdot 3^{-n}$, and the Euclidean distance is bounded by the path distance." You imply this, but spelling out the "shortest path" logic makes it watertight.

E. Complex Similitude Notation (Section 2.2) Defining the Iterated Function System on $\mathbb{C}$ using complex coefficients is standard, but you might want to briefly confirm for the reader that $S_2(z) = \frac{1}{2} + \frac{i\sqrt{3}}{6} + \frac{\bar{\omega}}{3}z$ is an **orientation-preserving** similitude (a rotation by $-\pi/3$), despite the presence of $\bar{\omega}$. To an undergraduate, the bar over ω might mistakenly trigger the thought of a geometric reflection (like $z \mapsto \bar{z}$). Clarifying that it's just multiplication by the constant $e^{-i\pi/3}$ prevents this pitfall.

3. Code Scrutiny

Your Python code is tight, pythonic, and numerically robust.

- Your use of `np.where` for the vectorized IFS is correct, though in modern NumPy, `np.select` or advanced indexing is slightly more readable for 4+ conditions. However, for a script appended to a mathematical essay, `np.where` is perfectly fine and requires no extra imports.
- The `koch_param` function evaluating the exact limit via base-4 expansion is a fantastic inclusion. Shrinking distances by 3^{-40} yields $\approx 8 \times 10^{-20}$, which as you rightly note, pushes the residual error beyond `float64` limits.

Final Verdict

This is publication-quality mathematical exposition. The concept of utilizing a unified Python script to verify fixed-point convergence, rough-path variation thresholds, and Hadamard gaps on a fractal, and then writing a paper around it, is pedagogical gold. Make the minor tweaks to the Fourier and Hölder explanations, and you have a flawless essay.